

Evaluating Scene-based in-situ Item Labeling for Immersive Conversational Recommendation

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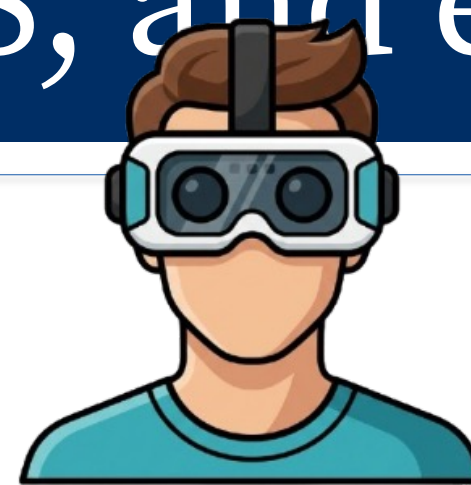
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TL;DR: We formalize immersive conversational recommendation systems (ICRS) as in-situ item labeling, define key information needs in ICRS, and evaluate how well LLMs, VLMs, and retrieval-based methods satisfy these needs.

1 Introduction

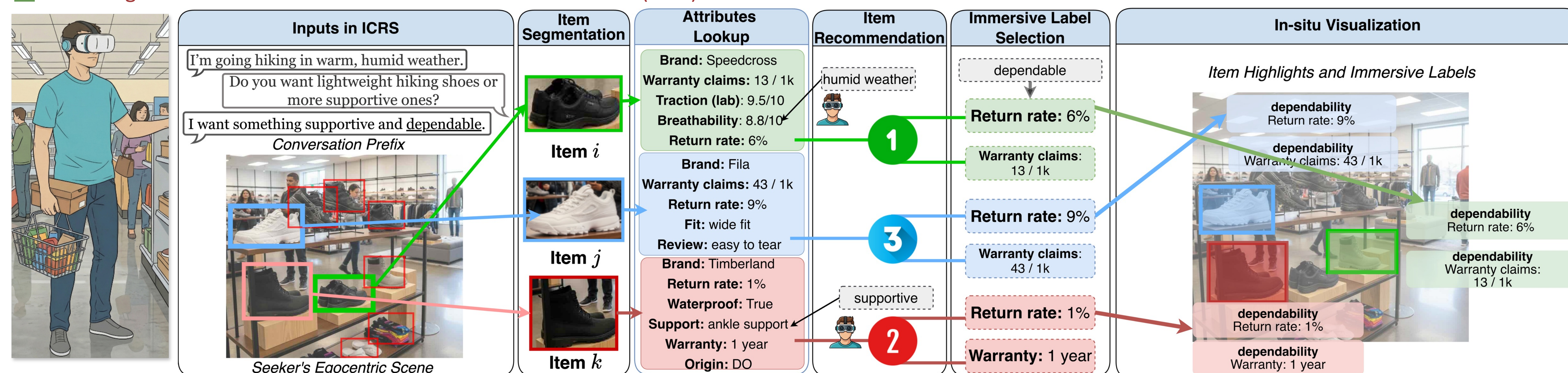


Transforming CRS from NL conversation alone to a POV scene that user can see.

Extended Reality enables recommendation directly within a seeker's egocentric scene. We formalize Immersive CRS (ICRS), where recommended items are highlighted in-view and augmented with concise contextual labels.

- 01 Formalize Immersive CRS and scene-based in-situ item labeling.
- 02 Define three label-selection criteria: RIN, PIN, and EIN.
- 03 Adapt three datasets; synthesize egocentric scenes for ICRS evaluation.
- 04 Benchmark retrieval and VLM-based methods and surface core limitations.

2 Six Stages of the Immersive Conversation Recommendation (ICRS)



what information should immersive labels present in user POV?

3 Examples Usage of ICRS

SCENE 01 Fashion
VOGUE · SHOES & APPAREL

Decisions fuse visual cues with non-visual info (reviews, fit, function).

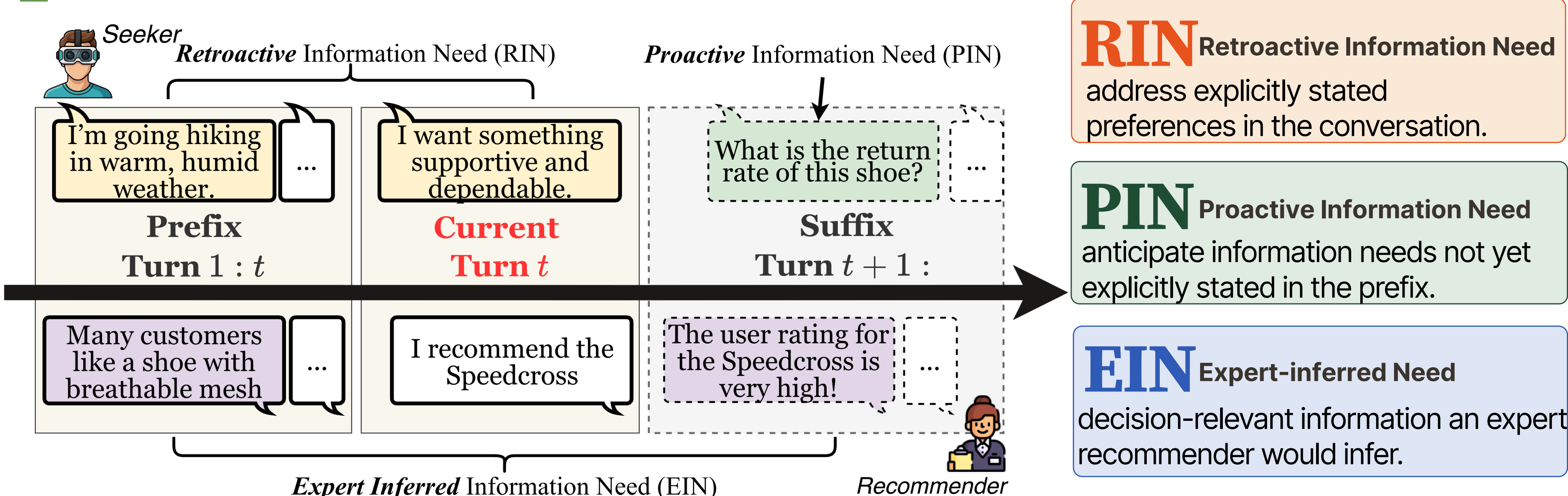
SCENE 02 Retail
COSREC · AMAZON PRODUCT

Items on the same shelf look identical, only metadata separates them.

SCENE 03 Movie
REDIAL · INSPIRED · TMDB

Posters already encode title & cast; labels must spend on the latent rest.

4 Three Levels of Information Needs

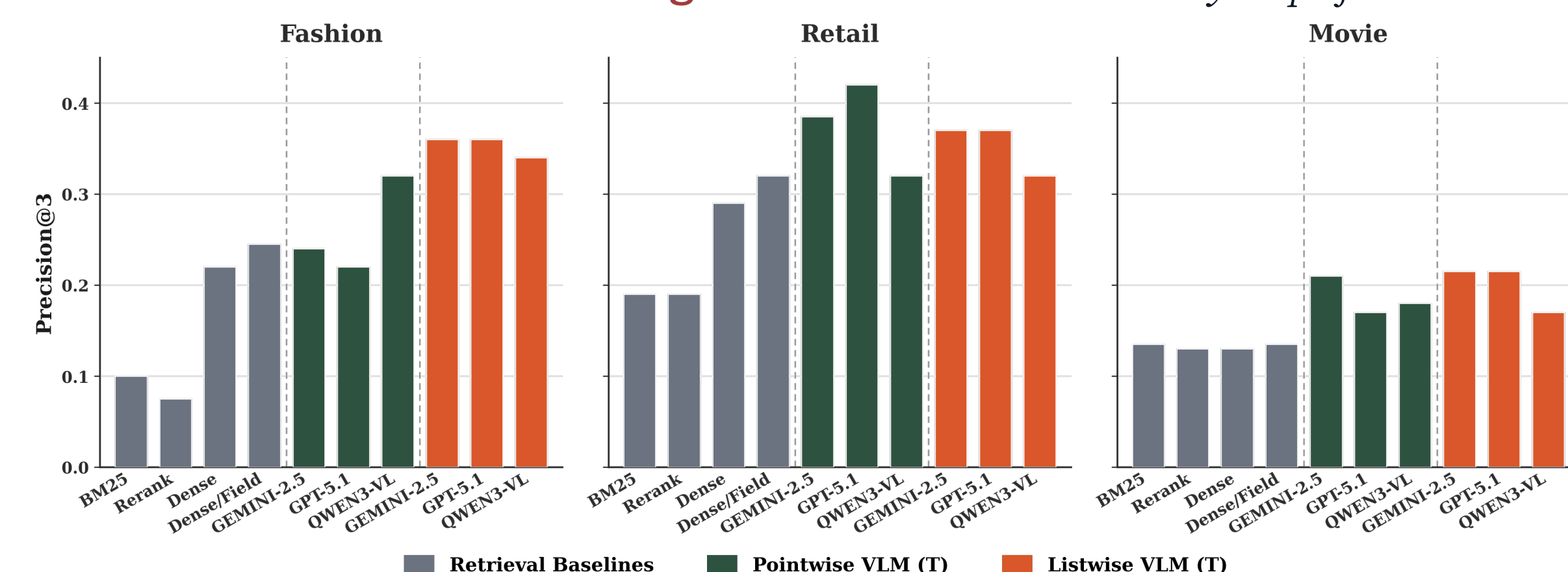


5 Per-criterion Labels Ground Truth and Evaluations

Intent Tags	Conversation Prefix $C_{1:t}$	Atomic Attributes
Retroactive Seeker Request	I'm going hiking in warm, humid weather.	Desc.: Engineered Mesh & Knit Uppers Color: brown
Express explicit preference	Many customers like a shoe with breathable mesh	Review: The design is great for ventilation
Proactive Seeker Request	I want something supportive and dependable.	Avg rating: 4.6/5
Reveal information needs not stated in prefix	I recommend Speedcross	Review: The lightweight mesh is comfortable
Expert Explanation	The user rating is very high!	Return rate: 6%
Justification or inference on seeker's info needs	What is the return rate of this shoe?	Avg rating: 9k reviews

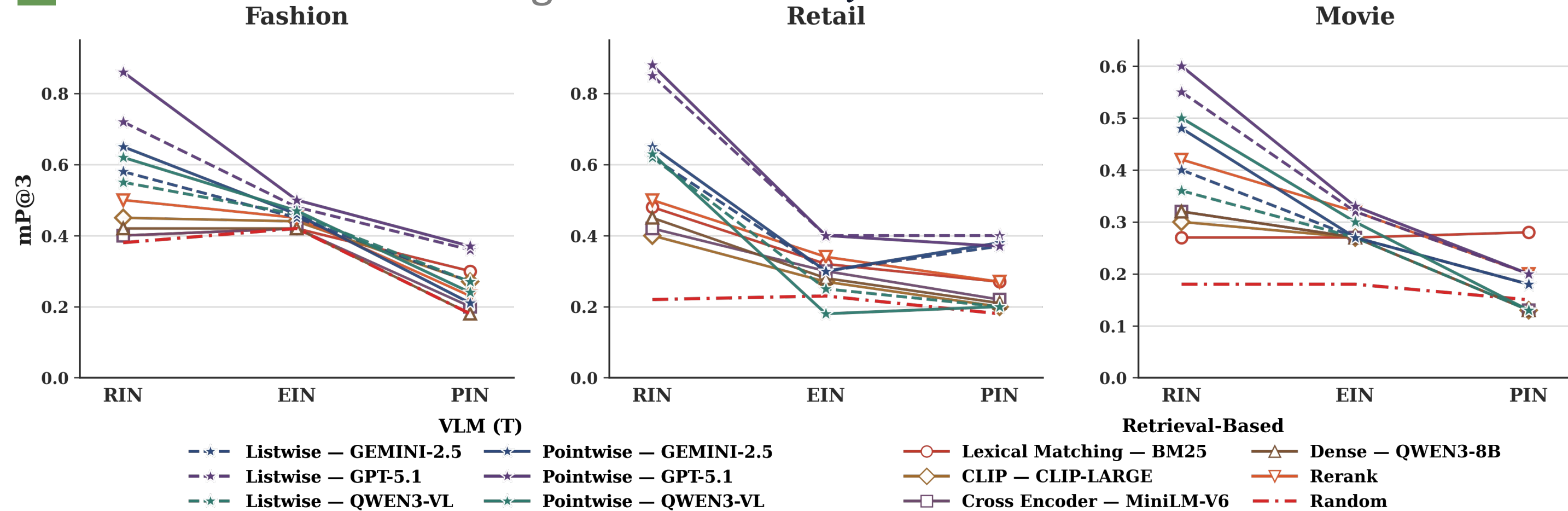
6 Item Recommendation Insight

Zero-shot VLMs consistently outperform retrieval-based CRS.



7 Immersive Labels Insight

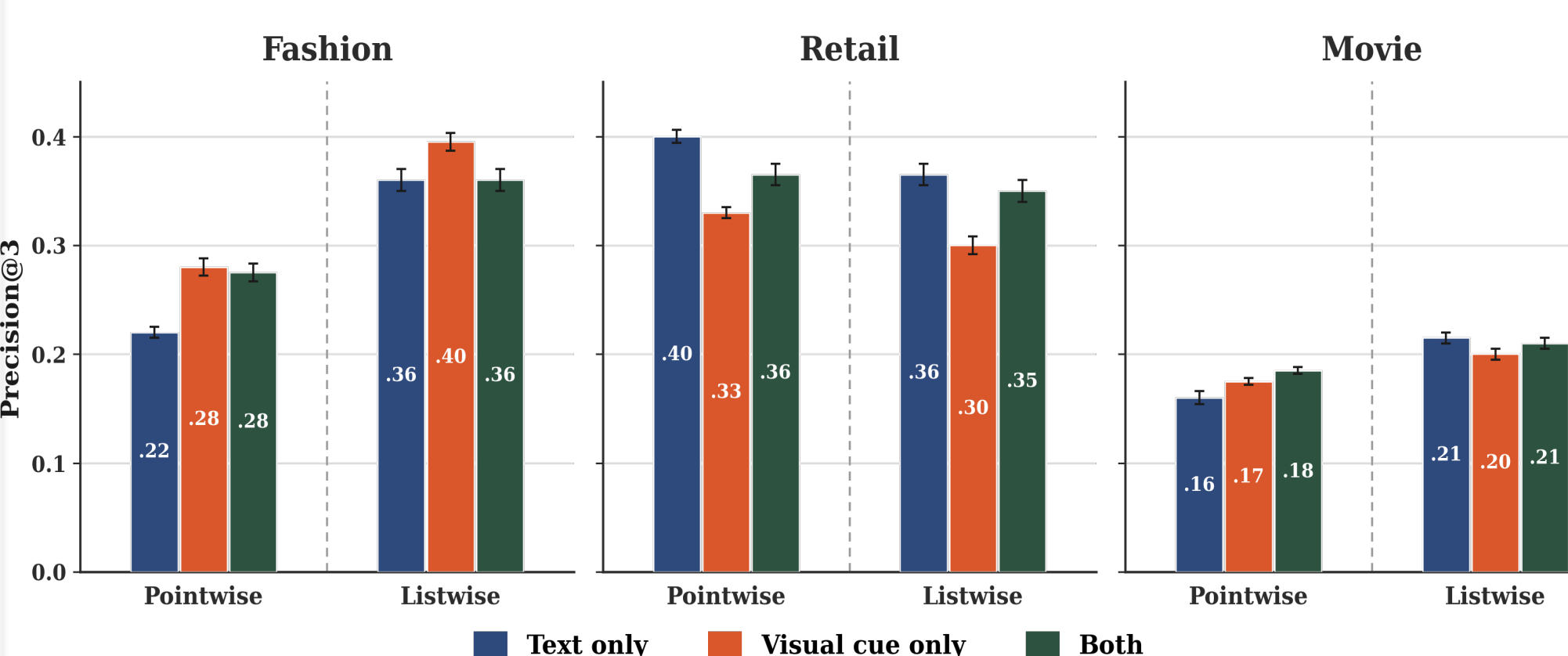
RIN is mostly solved; PIN and EIN remain hard.



LIMITATION · 01

Naive Feature Combination Can Hurt Performance

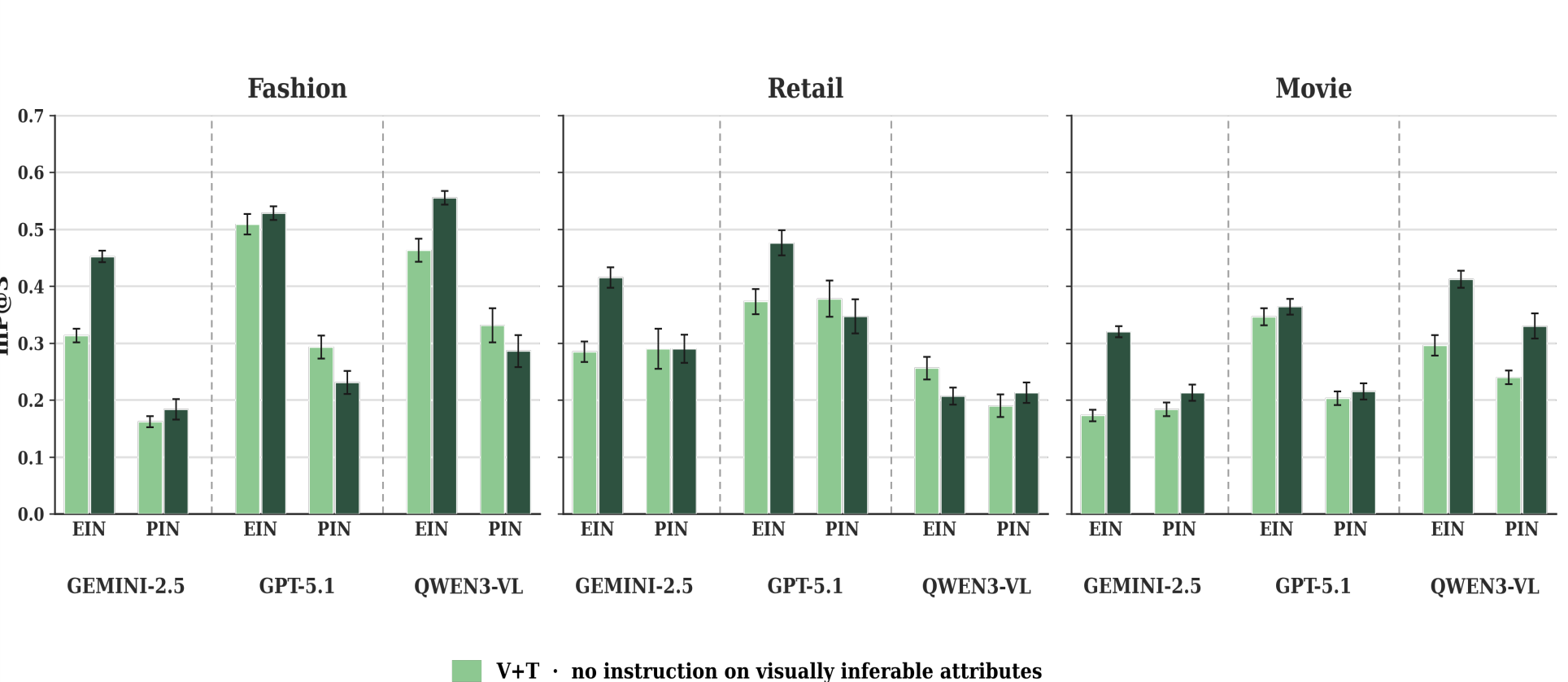
- Existing methods cannot effectively leverage the item modalities relevant to each scenario.
- Naively combining visual and textual features (V+T) can be inefficient and may even degrade performance.



LIMITATION · 02

Limited Latent Awareness

- Existing methods struggle to recognize latent information needs
- resulting in redundant presentation of visually inferable information.



LIMITATION · 03

Proactive Disclosure is the Hardest Challenge

Errors by domain:

- Movie:** visually inferable → poster reveals information
- Fashion:** explicit response → LLM does not aware of proactive needs
- Retail:** misaligned proactive → proactive is hard task

